

Paper 1	
Q1	$ax + \left(2a - \frac{a^2}{2}\right)x^2 + \left(\frac{a^3}{3} - a^2 + 2a\right)x^3 + \dots; a = 4$
Q2	(ii) $x < a$ or $x > a + \frac{1}{\sqrt{b}}$
Q3	(i) $x = -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ (ii) maximum
Q4	(ii) $y = 4 + \frac{1}{x+2}$; Asymptotes are $y = 4$ and $x = -2$
Q5	(i) $a = -1.5, b = 1.5, c = 7$ (ii) $x = -1.33$ (iii) $x = 1.15$ or -0.145
Q6	(iii) $\tilde{r} = \tilde{a} + \left(\frac{\tilde{d} - \tilde{a} \cdot \tilde{n}}{\tilde{b} \cdot \tilde{n}}\right)\tilde{b}$
Q7	(i) $\frac{\sin 2(m-n)x}{4(m-n)} - \frac{\sin 2(m+n)x}{4(m+n)} + C$ (ii) π
Q8	(a) $z = -1 + 2i; 2 - i$ (bi) $p = -6, q = -66$ (bii) $(w^2 - 2w + 2)(w^2 - 4w + 29)$
Q9	(ai) $u_n = A(2n-1) + B$ (aii) $A = 3, B = -9$ (b) $\sum_{r=1}^n r^3 = \frac{n^2(n+1)^2}{4}$ (c) $\sum_{r=0}^{\infty} \frac{x^r}{r!} = e^x$
Q10	(i) $a = -\frac{22}{5}$ (iii) $\left(\frac{3}{2}, \frac{1}{2}, -1\right); \sqrt{2.5}$ metres
Q11	(ib) $v = 10t + 4$ (ii) $v = \frac{10}{k}(1 - e^{-kt})$ (iii) 9.21 s

Paper 2	
Q1	(i) $2\sqrt{15}$ units (ii) $y = \frac{6}{x}$.
Q2	(i) $d = \frac{3}{2}$ (ii) $r = -1.4511$ or 1.2100 (iii) smallest value of n is 42
Q3	(ai) $\left(\frac{a}{2}, 0\right); (0, b)$ (aii) $(a+1, 0)$ (aiii) $\left(\frac{a+1}{2}, 0\right)$ (aiv) $(0, a); (b, 0)$ (bi) $a = 1$ (bii) $g^2(x) = x; g^{-1}(x) = \frac{x}{x-1}$ (biii) $b = 0$ or 2
Q4	(a) 15.1875 or $\frac{243}{16}$ units ² (bi) $\frac{\pi}{2a(a-1)}$ units ³ (bii) $b = \frac{1 + \sqrt{a^2 - a + 1}}{2}$
Q5	(ii) $E(T) = \frac{20}{7}; \text{Var}(T) = \frac{75}{98}$ (iii) 0.238
Q6	(i) 955514880 (ii) 1567641600 (iii) $\frac{1001}{3876}$ (or 0.258)
Q7	(ii) $\bar{x} = 31.8075; s^2 = 0.245$ (iii) p -value = 0.0140
Q8	(ii) $r = 0.981$
Q9	(ii) 0.632 (iii) 0.000104 (iv) 0.458 (vi) 0.421 (vii) 0.9408
Q10	(i) 0.345 (ii) 0.612 (iii) $\mu = 12.3; \sigma = 0.475$ (iv) $k = 55.7$

